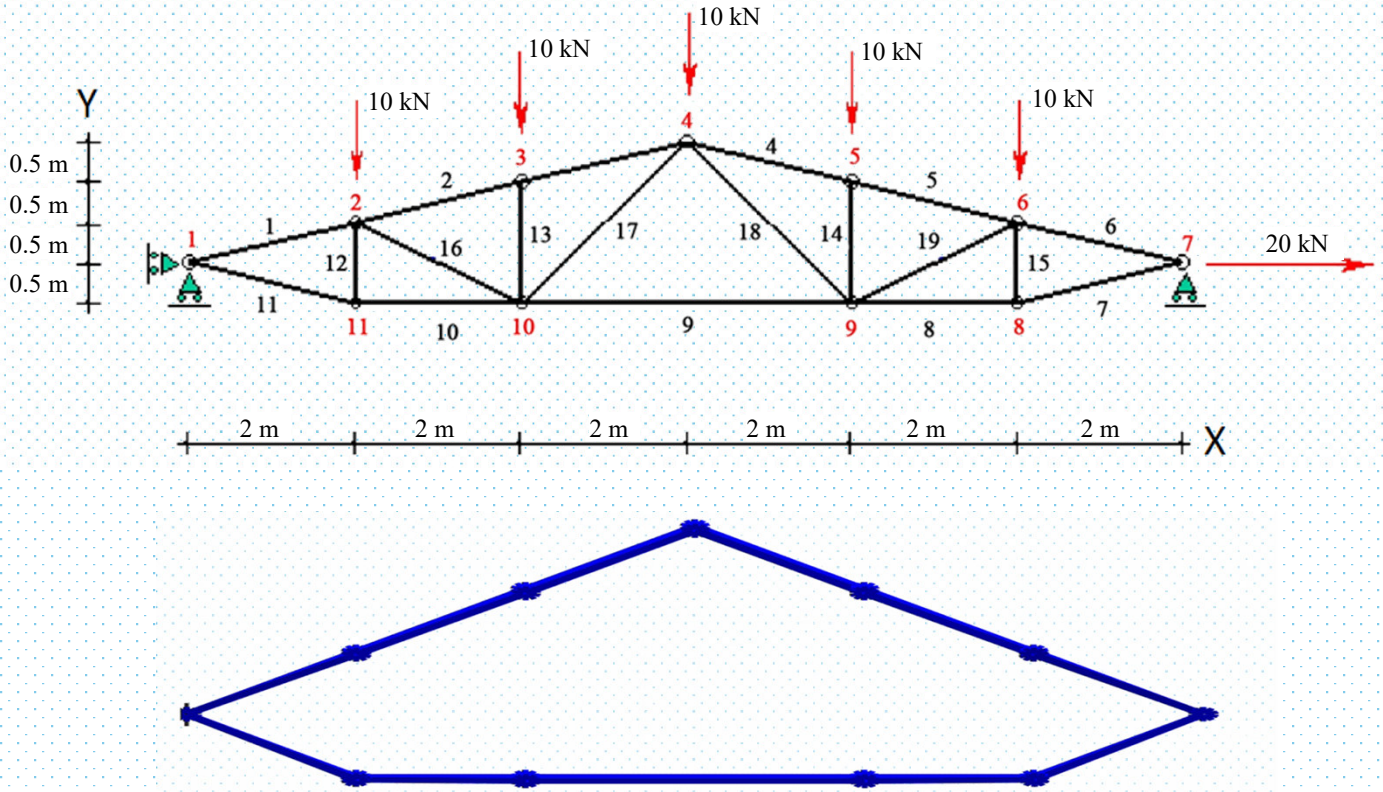


Bauman Moscow State Technical University

Special Topics in Finite Element Modeling for IC engines

Stress-Strain Analysis of a Plane Truss



Teachers: Malastowski Nikolay, Blinov Artem

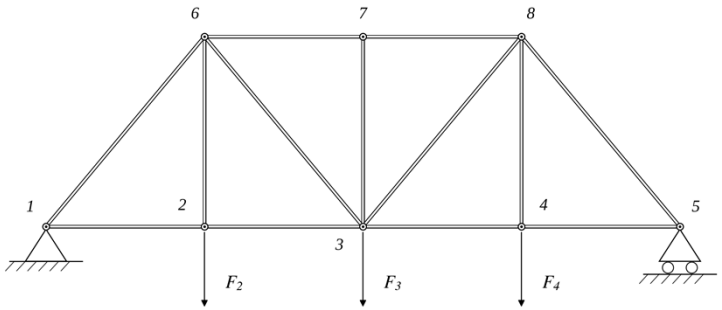
A truss and its plane structural model

A truss is a framework that remains geometrically stable
when rigid member connections are replaced by ideal pins.

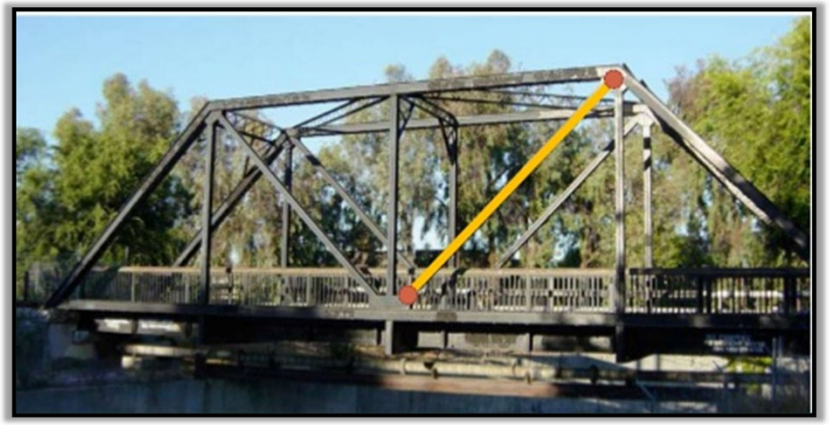
Plane truss

Its member axes and applied loads lie in a single plane.

Structural model of the side truss



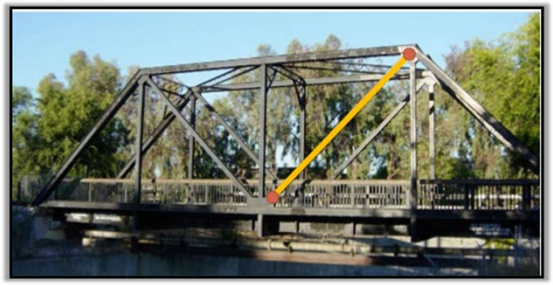
Actual structure



Pin joints and nodal loads produce axial member forces.
The bridge's out-of-plane bracing is considered separately.

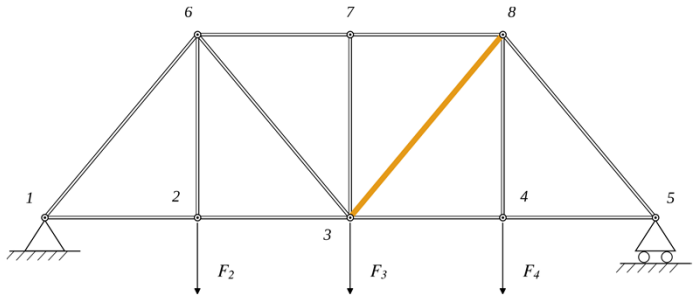
Truss: a naturally discrete structural model

Actual truss bridge



Bars are finite elements
Joints are nodes

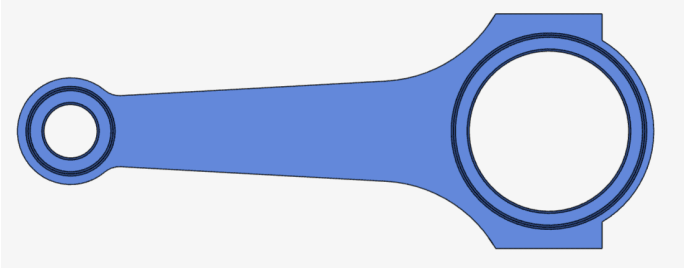
Pin-jointed bar model



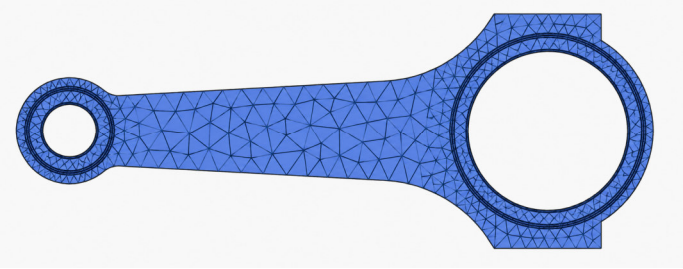
The truss itself defines a finite number of members and nodes.

Connecting rod: a continuum and its discrete finite element model

Connecting rod: geometric model



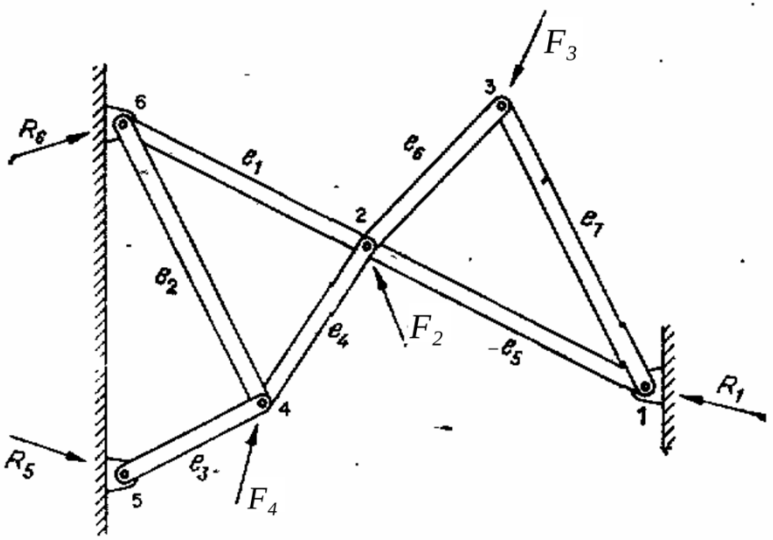
Connecting rod: finite element model



We divide the continuum into a finite number of elements connected at nodes.

Both use element force–displacement relations, nodal compatibility, and system equilibrium.

Problem statement and modeling assumptions



F: applied loads; R: support reactions.

Input data

Geometry, material properties, external loads, and support conditions

Unknown quantities

Nodal (joint) displacements
Member strains
Member stresses
Support reactions R

Modeling assumptions

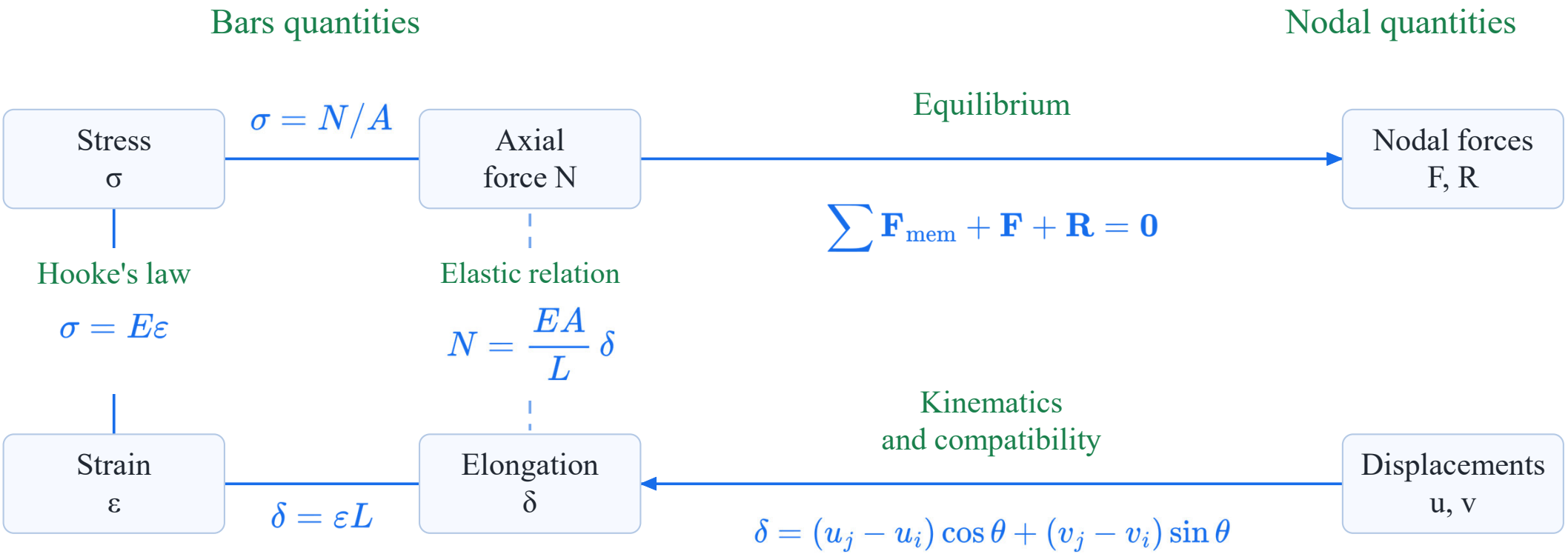
Bars and joints

Straight members, ideal pin joints,
Nodal loads. Tension and compression only.
Bending and buckling are not considered.

Linear static analysis

Small displacements and Hooke's law.
E and A are constant along each member.
No initial stresses or thermal strains.

Displacement, strain, and force relations in a plane truss

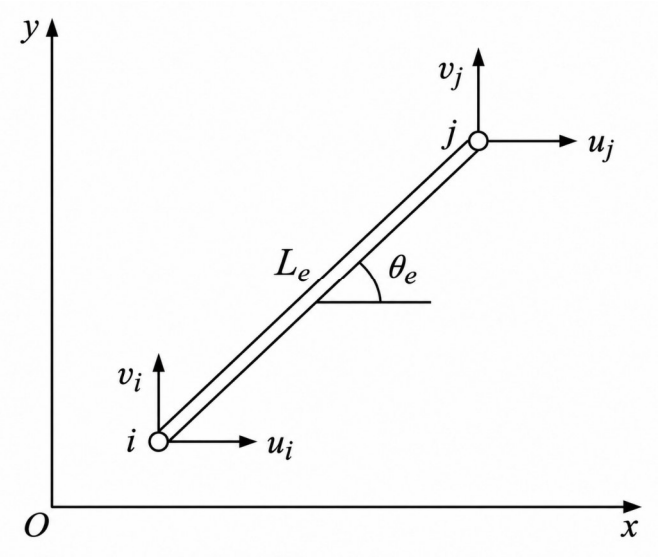


A: cross-sectional area; E: Young's modulus; L: length; θ: angle of i → j to x; i, j: member ends.

F: applied loads; R: unknown reactions; Fmem: member forces on a node; u, v: x- and y-displacements.

All relations hold simultaneously for the same equilibrium state of the truss.

Bar element: geometry, nodal displacements, and elongation



1. Unknown displacement components

$$\mathbf{U}_e = (u_i \; v_i \; u_j \; v_j)^{\mathsf{T}}$$

u: displacement in x; v: displacement in y.

2. Bar geometry: length and orientation

$$\cos\theta_e = \frac{x_j - x_i}{L_e}, \qquad \sin\theta_e = \frac{y_j - y_i}{L_e}$$

$$L_e = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$$

θ_e: angle of i → j to +x; L_e: initial length; e: member (finite element) number.

3. Bar elongation

$$(L_e + \Delta L_e)^2 = \left[(x_j - x_i) + (u_j - u_i)\right]^2 + \left[(y_j - y_i) + (v_j - v_i)\right]^2$$

For small displacements, expand the squares and neglect second-order terms:

$$\Delta L_e \approx (u_j - u_i)\cos\theta_e + (v_j - v_i)\sin\theta_e$$

Elongation, axial strain, stress, and force

1. Axial strain: relative change in length

$$\varepsilon_e = \frac{\Delta L_e}{L_e} = \frac{(u_j - u_i)\cos\theta_e + (v_j - v_i)\sin\theta_e}{L_e}$$

$$\mathbf{B}_e = \frac{1}{L_e}(-\cos\theta_e \quad -\sin\theta_e \quad \cos\theta_e \quad \sin\theta_e), \qquad \varepsilon_e = \mathbf{B}_e\mathbf{U}_e$$

B_e is the strain–displacement matrix relating nodal displacements to axial strain.

2. Stress: Hooke's law for a linear elastic member

$$\sigma_e = E_e\varepsilon_e$$

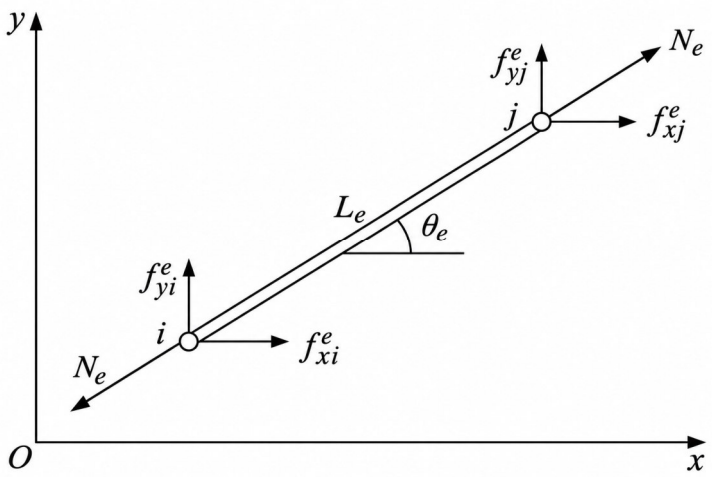
E_e is Young's modulus of the member material.

3. Axial force: stress × cross-sectional area A_e

$$N_e = A_e\sigma_e = \frac{E_eA_e}{L_e}\Delta L_e = \frac{E_eA_e}{L_e}\left[(u_j - u_i)\cos\theta_e + (v_j - v_i)\sin\theta_e\right]$$

ε_e is dimensionless; σ_e is in Pa and N_e in N. Positive ε_e, σ_e, N_e denote tension; negative values denote compression.

Nodal forces of a bar element



Nodal forces in terms of axial force

$$\mathbf{f}_e = \begin{pmatrix} f^e_{xi} \\ f^e_{yi} \\ f^e_{xj} \\ f^e_{yj} \end{pmatrix} = N_e \begin{pmatrix} -\cos\theta_e \\ -\sin\theta_e \\ \cos\theta_e \\ \sin\theta_e \end{pmatrix}$$

$$f^e_{xi} = \frac{E_e A_e}{L_e}\left[\cos^2\theta_e\, u_i + \sin\theta_e \cos\theta_e\, v_i - \cos^2\theta_e\, u_j - \sin\theta_e \cos\theta_e\, v_j\right]$$

$$f^e_{yi} = \frac{E_e A_e}{L_e}\left[\sin\theta_e \cos\theta_e\, u_i + \sin^2\theta_e\, v_i - \sin\theta_e \cos\theta_e\, u_j - \sin^2\theta_e\, v_j\right]$$

$$f^e_{xj} = \frac{E_e A_e}{L_e}\left[-\cos^2\theta_e\, u_i - \sin\theta_e \cos\theta_e\, v_i + \cos^2\theta_e\, u_j + \sin\theta_e \cos\theta_e\, v_j\right]$$

$$f^e_{yj} = \frac{E_e A_e}{L_e}\left[-\sin\theta_e \cos\theta_e\, u_i - \sin^2\theta_e\, v_i + \sin\theta_e \cos\theta_e\, u_j + \sin^2\theta_e\, v_j\right]$$

Nodal forces in terms of nodal displacements

Member stiffness and nodal equilibrium

Matrix form of the nodal force–displacement relation

$$\mathbf{f}_e = \mathbf{k}_e \mathbf{U}_e$$

$$\mathbf{k}_e = \frac{E_e A_e}{L_e} \begin{pmatrix} \cos^2\theta_e & \sin\theta_e\cos\theta_e & -\cos^2\theta_e & -\sin\theta_e\cos\theta_e \\ \sin\theta_e\cos\theta_e & \sin^2\theta_e & -\sin\theta_e\cos\theta_e & -\sin^2\theta_e \\ -\cos^2\theta_e & -\sin\theta_e\cos\theta_e & \cos^2\theta_e & \sin\theta_e\cos\theta_e \\ -\sin\theta_e\cos\theta_e & -\sin^2\theta_e & \sin\theta_e\cos\theta_e & \sin^2\theta_e \end{pmatrix}$$

Equilibrium at node a: member contributions in x and y

$$\sum_{e \ni a} f_{xa}^{(e)} = F_{xa} + R_{xa}, \qquad \sum_{e \ni a} f_{ya}^{(e)} = F_{ya} + R_{ya}$$

Sum only members connected to node a. F_a is the applied load and R_a the support reaction.

Stiffness relates unknown displacements to element forces, but does not by itself determine U_e.

System displacements and the member equation

1. Global displacement vector: two components per node

$$\mathbf{U} = (u_1\ v_1\ \cdots\ u_i\ v_i\ \cdots\ u_j\ v_j\ \cdots\ u_n\ v_n)^{\mathsf{T}}$$

$$U_{2i-1} = u_i, \quad U_{2i} = v_i, \qquad U_{2j-1} = u_j, \quad U_{2j} = v_j$$

Odd indices denote x-displacements and even indices y-displacements. n is the number of nodes.

2. Four scalar equations for member e with end nodes i and j

$$\begin{pmatrix} f_{xi}^{(e)} \\ f_{yi}^{(e)} \\ f_{xj}^{(e)} \\ f_{yj}^{(e)} \end{pmatrix} = \begin{pmatrix} k_{2i-1,2i-1}^{(e)} & k_{2i-1,2i}^{(e)} & k_{2i-1,2j-1}^{(e)} & k_{2i-1,2j}^{(e)} \\ k_{2i,2i-1}^{(e)} & k_{2i,2i}^{(e)} & k_{2i,2j-1}^{(e)} & k_{2i,2j}^{(e)} \\ k_{2j-1,2i-1}^{(e)} & k_{2j-1,2i}^{(e)} & k_{2j-1,2j-1}^{(e)} & k_{2j-1,2j}^{(e)} \\ k_{2j,2i-1}^{(e)} & k_{2j,2i}^{(e)} & k_{2j,2j-1}^{(e)} & k_{2j,2j}^{(e)} \end{pmatrix} \begin{pmatrix} u_i \\ v_i \\ u_j \\ v_j \end{pmatrix}$$

Coefficient indices: first, the global force number; second, the global displacement number.

The member matrix remains 4 × 4, with row and column indices taken from the global system.

Expanding the member matrix and force vector

Expanded element stiffness matrix

$$\mathbf{K}^{(e)} = \begin{pmatrix} \vdots & 0 & 0 & \cdots & 0 & 0 & \vdots \\ 0 & k^{(e)}_{2i-1,2i-1} & k^{(e)}_{2i-1,2i} & 0 & k^{(e)}_{2i-1,2j-1} & k^{(e)}_{2i-1,2j} & 0 \\ 0 & k^{(e)}_{2i,2i-1} & k^{(e)}_{2i,2i} & 0 & k^{(e)}_{2i,2j-1} & k^{(e)}_{2i,2j} & 0 \\ \vdots & 0 & 0 & \cdots & 0 & 0 & \vdots \\ 0 & k^{(e)}_{2j-1,2i-1} & k^{(e)}_{2j-1,2i} & 0 & k^{(e)}_{2j-1,2j-1} & k^{(e)}_{2j-1,2j} & 0 \\ 0 & k^{(e)}_{2j,2i-1} & k^{(e)}_{2j,2i} & 0 & k^{(e)}_{2j,2j-1} & k^{(e)}_{2j,2j} & 0 \\ \vdots & 0 & 0 & \cdots & 0 & 0 & \vdots \end{pmatrix}_{2n \times 2n}$$

Expanded element nodal force vector

$$\bar{\mathbf{f}}^{(e)} = \begin{pmatrix} 0 & \cdots \\ f^{(e)}_{xi} & 2i-1 \\ f^{(e)}_{yi} & 2i \\ 0 & \cdots \\ f^{(e)}_{xj} & 2j-1 \\ f^{(e)}_{yj} & 2j \\ 0 & \cdots \end{pmatrix}_{2n \times 1}$$

Expanded element matrix equation

$$\bar{\mathbf{f}}^{(e)} = \mathbf{K}^{(e)}\mathbf{U}$$

Active indices: 2i − 1, 2i, 2j − 1, 2j. Ellipses denote the remaining zero rows and columns.

Vector: up to 4 nonzero components. Matrix: up to 16 nonzero coefficients.

Assembly of the global truss equations

1. Summing element contributions to the global system

$$\sum_{e=1}^{m} \bar{\mathbf{f}}^{(e)} = \left(\sum_{e=1}^{m} \mathbf{K}^{(e)} \right) \mathbf{U} = \mathbf{K}\mathbf{U} = \mathbf{F} + \mathbf{R}$$

F: applied external loads; R: unknown support reactions; m: number of members.
The components of R associated with free degrees of freedom are zero.

2. Scalar form of the global system K U = F + R

$$\begin{pmatrix} K_{1,1} & K_{1,2} & \cdots & K_{1,2n-1} & K_{1,2n} \\ K_{2,1} & K_{2,2} & \cdots & K_{2,2n-1} & K_{2,2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ K_{2n-1,1} & K_{2n-1,2} & \cdots & K_{2n-1,2n-1} & K_{2n-1,2n} \\ K_{2n,1} & K_{2n,2} & \cdots & K_{2n,2n-1} & K_{2n,2n} \end{pmatrix} \begin{pmatrix} u_1 \\ v_1 \\ \vdots \\ u_n \\ v_n \end{pmatrix} = \begin{pmatrix} F_{x1} + R_{x1} \\ F_{y1} + R_{y1} \\ \vdots \\ F_{xn} + R_{xn} \\ F_{yn} + R_{yn} \end{pmatrix}$$

K_{11}, K_{12}, … sum the member contributions at the corresponding global matrix positions.

Each node has two equilibrium equations: one in x and one in y.

Boundary conditions and solution of the linear system

1. Applying boundary conditions

Kinematic: prescribe known displacements

Force: specify the external loads F

Global matrix equation

$$\begin{pmatrix} K_{1,1} & K_{1,2} & K_{1,3} & \cdots & K_{1,2n-2} & K_{1,2n-1} & K_{1,2n} \\ K_{2,1} & K_{2,2} & K_{2,3} & \cdots & K_{2,2n-2} & K_{2,2n-1} & K_{2,2n} \\ K_{3,1} & K_{3,2} & K_{3,3} & \cdots & K_{3,2n-2} & K_{3,2n-1} & K_{3,2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ K_{2n-2,1} & K_{2n-2,2} & K_{2n-2,3} & \cdots & K_{2n-2,2n-2} & K_{2n-2,2n-1} & K_{2n-2,2n} \\ K_{2n-1,1} & K_{2n-1,2} & K_{2n-1,3} & \cdots & K_{2n-1,2n-2} & K_{2n-1,2n-1} & K_{2n-1,2n} \\ K_{2n,1} & K_{2n,2} & K_{2n,3} & \cdots & K_{2n,2n-2} & K_{2n,2n-1} & K_{2n,2n} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ u_2 \\ \vdots \\ v_{n-1} \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} F_1 + R_1 \\ F_2 + R_2 \\ F_3 \\ \vdots \\ F_{2n-2} \\ F_{2n-1} + R_{2n-1} \\ F_{2n} + R_{2n} \end{pmatrix}$$

Matrix equation with support constraints

$$\Longrightarrow \begin{pmatrix} K_{3,3} & \cdots & K_{3,2n-2} \\ \vdots & \ddots & \vdots \\ K_{2n-2,3} & \cdots & K_{2n-2,2n-2} \end{pmatrix} \begin{pmatrix} u_2 \\ \vdots \\ v_{n-1} \end{pmatrix} = \begin{pmatrix} F_3 \\ \vdots \\ F_{2n-2} \end{pmatrix}$$

For prescribed zero displacements, remove the corresponding rows and columns; R = 0 at free DOFs.

2. Solving the linear system

Determine the free displacements:

$$(u_2\ v_2\ \cdots\ u_{n-1}\ v_{n-1})^{\mathsf{T}}$$

3. Recovering the full vector U

Restore the prescribed zero displacements.

$$\mathbf{U} = (0\ 0\ u_2\ v_2\ \cdots\ u_{n-1}\ v_{n-1}\ 0\ 0)^{\mathsf{T}}$$

Solve the reduced system for the unknown displacements, then restore prescribed values to obtain the full vector U

Support reactions and equilibrium checks

1. Determining support reactions

$$\mathbf{R} = \mathbf{KU} - \mathbf{F}$$

K is the original full matrix; U contains the computed and prescribed displacements.

$$R_{xa} = \sum_{b=1}^{n} \left(K_{2a-1,\,2b-1} u_b + K_{2a-1,\,2b} v_b \right) - F_{xa}$$

$$R_{ya} = \sum_{b=1}^{n} \left(K_{2a,\,2b-1} u_b + K_{2a,\,2b} v_b \right) - F_{ya}$$

a, b: node numbers; n: number of nodes. Reactions act in constrained directions.

2. Checking nodal equilibrium

In free directions, the corresponding components of KU − F ≈ 0.

3. Checking equilibrium of the whole truss

$$\sum_{a=1}^{n} (F_{xa} + R_{xa}) = 0, \qquad \sum_{a=1}^{n} (F_{ya} + R_{ya}) = 0$$

The sums of applied external forces and support reactions must be zero in both x and y.

Member strains, stresses, and axial forces

1. Extract the computed displacements at the member ends

$$\mathbf{U} = (u_1\ v_1\ \cdots\ u_n\ v_n)^{\mathsf{T}} \quad \longrightarrow \quad \mathbf{U}_e = (u_i\ v_i\ u_j\ v_j)^{\mathsf{T}}$$

e: member number; i, j: end nodes. All components of U are now known.

2. Find elongation from the relative end displacements

$$\Delta L_e = (u_j - u_i)\cos\theta_e + (v_j - v_i)\sin\theta_e$$

L_e: initial length; θ_e: angle of i → j to the positive x-direction.

3. Compute strain, stress, and axial force

$$\varepsilon_e = \frac{\Delta L_e}{L_e}, \qquad \sigma_e = E_e\varepsilon_e, \qquad N_e = A_e\sigma_e$$

E_e: Young's modulus; A_e: cross-sectional area. $N_e > 0$ denotes tension; $N_e < 0$ denotes compression.

No new linear system is needed: use the computed displacements to find ΔL_e, ε_e, σ_e, and N_e in turn.

General procedure for plane truss analysis

1. Specify geometry, material, loads, and supports; U remains unknown.

2. Express each member's elongation and force in terms of the unknown U.

3. Enforce nodal compatibility and write the nodal equilibrium equations.

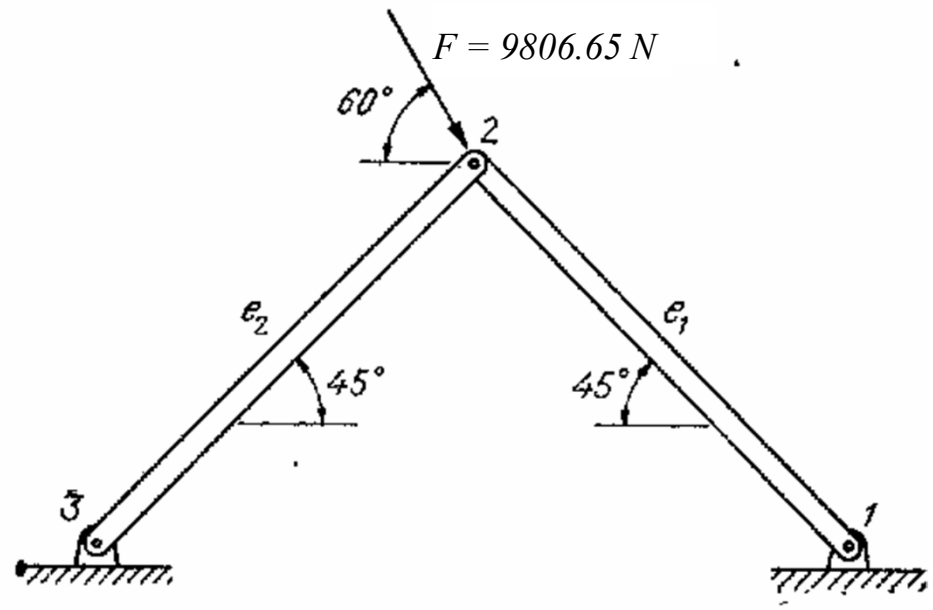
4. Apply prescribed displacements and solve for the unknown components of U

5. Find reactions, then ΔL, ε, σ, and N for each member.

6. Check equilibrium, units, and the physical meaning of the results.

The general derivation is complete. Next, we apply it to a two-member system.

Two-bar truss: model and input data



1. Identical bars and fixed support nodes

$$L = 0.1\ \text{m}, \quad A = 10^{-4}\ \text{m}^2$$

$$E = 196.133 \cdot 10^{9}\ \text{Pa}$$

Nodes 1 and 3 are restrained in x and y.
The unknowns are u_2 and v_2 at node 2.

2. Bar orientations

$$e_1 = (1, 2), \quad \theta_1 = 135^\circ$$

$$e_2 = (2, 3), \quad \theta_2 = 225^\circ$$

3. Load components in the global x, y axes

$$F_{x2} = F\cos 60^\circ = 4903.325\ \text{N}, \quad F_{y2} = -F\sin 60^\circ = -8492.808\ \text{N}$$

F = 9806.65 N. The x-axis points right and the y-axis up; the force acts only at node 2.

Stiffness matrices of the two members

Right member: 1 → 2

$$\cos 135^\circ = -\frac{1}{\sqrt{2}}, \quad \sin 135^\circ = \frac{1}{\sqrt{2}}$$

$$\mathbf{k}_1 = \frac{EA}{2L}\begin{pmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

Order: u_1, v_1, u_2, v_2.

Left member: 2 → 3

$$\cos 225^\circ = -\frac{1}{\sqrt{2}}, \quad \sin 225^\circ = -\frac{1}{\sqrt{2}}$$

$$\mathbf{k}_2 = \frac{EA}{2L}\begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{pmatrix}$$

Order: u_2, v_2, u_3, v_3.

Common stiffness factor

$$\frac{EA}{2L} = 98\,066\,500 \text{ N/m}$$

Substitute the directions into the general member matrix; the signs of the coupling terms depend on orientation.

Expanding element matrices to 6 × 6

Element 1: rows and columns 1–4

$$\mathbf{K}^{(1)} = \frac{EA}{2L}\begin{pmatrix} 1 & -1 & -1 & 1 & 0 & 0 \\ -1 & 1 & 1 & -1 & 0 & 0 \\ -1 & 1 & 1 & -1 & 0 & 0 \\ 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Element 1 has no contribution at node 3.

Element 2: rows and columns 3–6

$$\mathbf{K}^{(2)} = \frac{EA}{2L}\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & -1 & -1 \\ 0 & 0 & 1 & 1 & -1 & -1 \\ 0 & 0 & -1 & -1 & 1 & 1 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{pmatrix}$$

Element 2 has no contribution at node 1.

The same displacement order for both elements

$$\mathbf{U} = (u_1\ v_1\ u_2\ v_2\ u_3\ v_3)^{\mathsf{T}}$$

$$\overline{\mathbf{f}}^{(1)} = \mathbf{K}^{(1)}\mathbf{U}, \qquad \overline{\mathbf{f}}^{(2)} = \mathbf{K}^{(2)}\mathbf{U}$$

Embed each original 4 × 4 matrix in a global 6 × 6 matrix; all remaining entries are zero.

Global assembly and support constraints

1. Add coefficients with the same global indices

$$\mathbf{K} = \mathbf{K}^{(1)} + \mathbf{K}^{(2)}, \qquad \mathbf{KU} = \mathbf{F} + \mathbf{R}$$

2. Prescribe $u_1 = v_1 = u_3 = v_3 = 0$

$$\frac{EA}{2L}\begin{pmatrix} 1 & -1 & -1 & 1 & 0 & 0 \\ -1 & 1 & 1 & -1 & 0 & 0 \\ -1 & 1 & 2 & 0 & -1 & -1 \\ 1 & -1 & 0 & 2 & -1 & -1 \\ 0 & 0 & -1 & -1 & 1 & 1 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 0 \\ u_2 \\ v_2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} R_{x1} \\ R_{y1} \\ 4903.325 \\ -8492.808 \\ R_{x3} \\ R_{y3} \end{pmatrix}$$

Row order: u_1, v_1, u_2, v_2, u_3, v_3. Both members contribute at node 2.

To solve for displacements, remove rows and columns 1, 2, 5, 6; retain 3 and 4.

Factor EA/(2L) = 98 066 500 N/m. Keep the original support rows to calculate reactions.

Reduced-system solution and the full vector U

1. Two equations for the free displacements at node 2

$$196\,133\,000 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_2 \\ v_2 \end{pmatrix} = \begin{pmatrix} 4903.325 \\ -8492.808 \end{pmatrix}$$

2. Solve for displacements

$$u_2 = \frac{4903.325}{196\,133\,000} = 2.5 \cdot 10^{-5}\ \text{m}$$

$$v_2 = \frac{-8492.808}{196\,133\,000} \approx -4.330127 \cdot 10^{-5}\ \text{m}$$

3. Restore the prescribed zeros

$$\mathbf{U} = (0 \quad 0 \quad 2.5 \cdot 10^{-5} \quad -4.330127 \cdot 10^{-5} \quad 0 \quad 0)^{\mathsf{T}}\ \text{m}$$

Node 2 moves 0.025 mm right and 0.043301 mm down; both support nodes remain stationary.

Support reactions and force-sum checks

1. Substitute the full U into the original support rows

$$R_{x1} = \frac{EA}{2L}(-u_2 + v_2), \quad R_{y1} = \frac{EA}{2L}(u_2 - v_2)$$

$$R_{x3} = R_{y3} = -\frac{EA}{2L}(u_2 + v_2)$$

$$R_{x1} = -6698.067\ \text{N}, \qquad R_{y1} = 6698.067\ \text{N}$$

$$R_{x3} = R_{y3} = 1794.742\ \text{N}$$

2. Check equilibrium of the whole truss in x and y

$$4903.325 - 6698.067 + 1794.742 \approx 0\ \text{N}$$

$$-8492.808 + 6698.067 + 1794.742 \approx 0\ \text{N}$$

The sums of loads and reactions are zero; the small residual in the displayed values is due to rounding.

Elongations, strains, stresses, and forces

1. Calculate elongations from the computed displacements

$$\Delta L_1 = \frac{-u_2 + v_2}{\sqrt{2}} = -4.829629 \cdot 10^{-5}\ \text{m}$$

$$\Delta L_2 = \frac{u_2 + v_2}{\sqrt{2}} = -1.294095 \cdot 10^{-5}\ \text{m}$$

2. Apply the geometric relation and Hooke's law

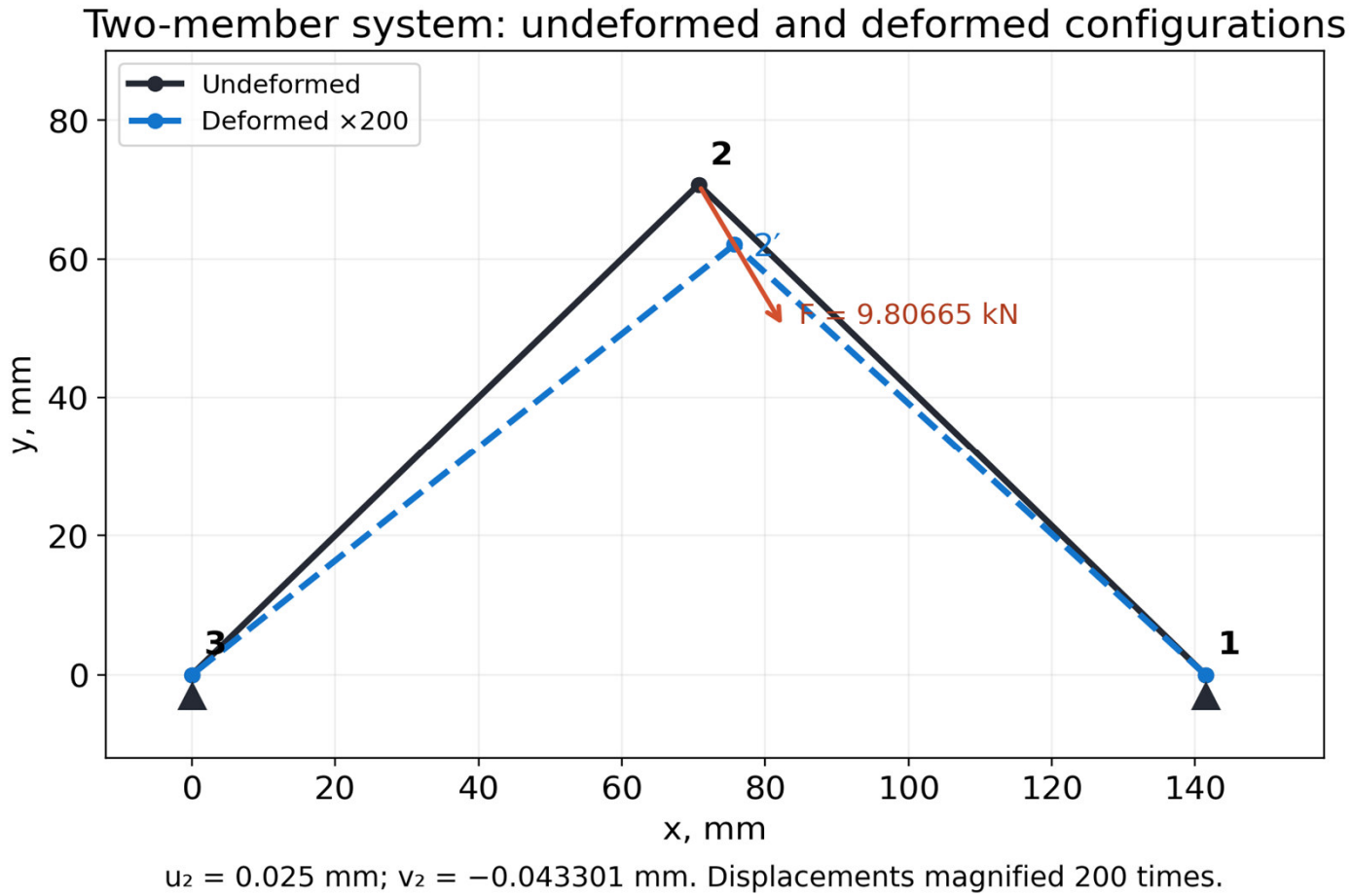
$$\varepsilon_e = \frac{\Delta L_e}{L}, \qquad \sigma_e = E\varepsilon_e, \qquad N_e = A\sigma_e$$

$$\varepsilon_1 = -4.829629 \cdot 10^{-4}, \quad \sigma_1 = -94.725\ \text{MPa}, \quad N_1 = -9472.497\ \text{N}$$

$$\varepsilon_2 = -1.294095 \cdot 10^{-4}, \quad \sigma_2 = -25.381\ \text{MPa}, \quad N_2 = -2538.148\ \text{N}$$

Both members shorten: ΔL, ε, σ, and N are negative. Both members are in compression.

Undeformed and deformed configurations



$u_2 = 0.025$ mm; $v_2 = -0.043301$ mm. Displacements magnified 200 times.

Displacements are magnified 200 times. The actual displacement of node 2 is much smaller.



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Special Topics in Finite Element Modeling for IC engines

CONCLUSION